Academic Technical Project Report — Business Intelligence & Geospatial Warehousing

Urban Intelligence CDMX: Geospatial Data Warehouse and Spatial Analysis of Urban Crime and Economic Density

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Abstract

This report documents the design, implementation, and econometric validation of an end-to-end Geospatial Data Warehouse (GDW) for Mexico City at the urban Basic Geostatistical Area (AGEB) grain. Implemented in PostgreSQL and PostGIS via Docker Compose, the dimensional model integrates 2,431 official AGEB polygons (accounting for 9.14 million residents from the INEGI 2020 Population Census), 472,617 active economic units from the National Directory of Economic Units (DENUE), and 196,990 geocoded crime records from the Mexico City Attorney General's Office (FGJ). Complete enumeration and topological alignment justify the selection of the 2020 Census over sample estimates. Thirteen territorial Key Performance Indicators are materialized. Diagnostic evaluations resolve the Small Number Problem in per-capita crime rates through area-normalized densities and crime-to-business ratios. Spatial econometric models utilizing row-standardized Queen contiguity weights demonstrate that crime density correlates significantly more strongly with business density (Pearson correlation of 0.50, Spearman rank correlation of 0.58) than with resident population density (Pearson correlation of 0.28, Spearman correlation of 0.36). Global Moran's I confirms intense spatial clustering (0.5568 for crime and 0.4371 for businesses, with pseudo p-values below 0.001), while Bivariate Moran's I (0.3432) and Local Indicators of Spatial Association (LISA) reveal that 91.6% of all commercial hot spots co-locate directly with high-crime clusters in the central urban corridors.

Index Terms

Geospatial Data Warehouse, PostgreSQL, PostGIS, Star Schema, Urban AGEB, Censo 2020, DENUE, Global Moran's I, LISA, Bivariate Moran's I, Mexico City.



I. Introduction and Scope

Metropolitan public safety and economic planning depend on the systematic integration of demographic, commercial, and criminal records produced by independent institutions. In Mexico City (CDMX), these datasets exhibit incompatible spatial representations: census data are reported as tabular aggregates, commercial units are registered as discrete points, and criminal offenses are logged as incident coordinates. Without a unified dimensional data warehouse, municipal evaluations remain vulnerable to spatial aggregation errors and irreproducible ad-hoc processing.

This project delivers a reproducible, containerized **Geospatial Data Warehouse (GDW)** for Mexico City based on the Unit 2 Project rubric specifications. The system unifies four foundational data sources at the urban *Área Geoestadística Básica* (AGEB) grain:

1) **Cartography:** Official Marco Geoestadístico 2020 polygons (2,431 urban AGEBs) reprojected to EPSG:6372 (Mexico ITRF2008 / LCC).
2) **Demographics:** INEGI 2020 Population and Housing Census (9,139,836 residents) [1].
3) **Economy:** INEGI National Directory of Economic Units (DENUE 2020), containing 472,617 active establishments across 937 SCIAN classes [2].
4) **Public Safety:** FGJ-CDMX criminal investigation records, containing 196,990 georeferenced incident events from the year 2020 [3].

Deployed on Docker Compose with PostgreSQL 15 and PostGIS 3.3 [4], the platform computes 13 territorial Key Performance Indicators (KPIs), resolves the Small Number Problem in per-capita crime rates, and executes spatial econometric models including Queen contiguity spatial weights, correlation matrices, Global Moran's I, Bivariate Moran's I, and Local Indicators of Spatial Association (LISA) [5], [6].

II. Methodological Foundation and Data Engineering

A. Selection of Analysis Grain: The Urban AGEB

Defining the grain of fact tables is the foundational step in dimensional modeling [7]. Candidate spatial units in Mexico City include municipalities (*alcaldías*, 16 units), postal codes (approximately 1,450 units), and urban AGEBs (2,431 units). Municipalities are excluded because their vast scale aggregates away critical neighborhood heterogeneities, inducing the Modifiable Areal Unit Problem (Openshaw). Postal codes are excluded because they are non-standard administrative boundaries that lack official demographic census tabulations.

The **Urban AGEB** (`cvegeo`, a standardized 13-character geographic identifier) is selected as the conformed spatial grain. Urban AGEBs represent the finest official geostatistical boundary with complete demographic coverage, bounded by recognizable urban infrastructure, and averaging 3,759 inhabitants per polygon.

TABLE I
Source Data Inventory and Extraction Profile

Layer	Source	Original Content & Grain	Raw Records
Geographic	INEGI Marco 2020	Vector polygons (`09a.shp`), EPSG:6372	2,431 AGEBs
Demographic	INEGI Censo 2020	Tabular census by AGEB, 222 socioeconomic variables	2,431 rows
Economic	INEGI DENUE 2020	Establishments with coordinates, SCIAN classification	472,617 units
Public Safety	FGJ-CDMX 2020	Criminal investigation files with coordinates, type, time	196,990 files

B. Methodological Defense of the 2020 Census Dataset

The selection of the **INEGI Censo 2020** [1] over sample-based estimates is grounded in three methodological requirements:

1) **Exhaustive Spatial Granularity:** The decennial census is the only demographic instrument in Mexico providing universal physical enumeration down to the individual AGEB. Sample surveys provide statistical validity only at municipal or state levels; downscaling them to neighborhoods requires synthetic imputation that introduces artificial variance.
2) **Topological Coherence:** Censo 2020 tabular attributes align deterministically, key-for-key, with the official 2020 *Marco Geoestadístico* cartographic boundary shapefiles (`09a.shp`). This guarantees an exact 1-to-1 cardinality without missing polygons, synthetic expansion factors, or boundary mismatches.
3) **Temporal Synchronization:** The 2020 baseline provides temporal coherence across demographics (Censo 2020), business operations (DENUE 2020), and crime registries (FGJ 2020), eliminating inter-annual distortion.

C. Cartographic Standards and Reprojection

To compute accurate geometric measurements (surface areas in square kilometers and Euclidean proximity buffers), all geometries are projected into **EPSG:6372** (*Mexico ITRF2008 / LCC*). Geographic coordinate systems (WGS84, EPSG:4326) measure space in angular degrees, distorting spatial densities. EPSG:6372 uses an official Lambert Conformal Conic projection, ensuring metric preservation across Mexico City.

III. Geospatial Warehouse Architecture

A. Containerized Infrastructure

The data warehouse is provisioned via `docker-compose.yml` using the official `postgis/postgis:15-3.3` image [4] with `platform: linux/amd64`. The container maintains automated health checks, persists storage volumes, and isolates the database environment across operating systems.

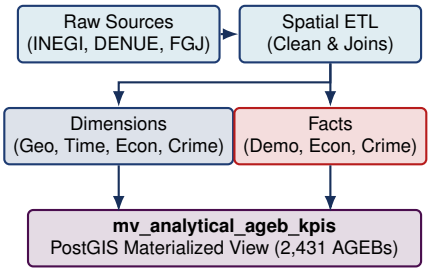


Fig. 1. Dimensional Data Warehouse ETL and materialized view architecture.

B. Multidimensional Star Schema

The relational model implements a Kimball-style multidimensional architecture [7] with conformed dimensions and three granular fact tables:

1) Conformed Dimensions

- `dim_geography`: 2,431 rows. Primary key `cvegeo`. Stores municipality names, polygon surface area in square kilometers (`area_km2`), and MultiPolygon geometry (`EPSG:6372`).
- `dim_time`: 366 rows (leap year 2020). Primary key `date_id`. Models calendar hierarchies (year, quarter, month, day, weekday).
- `dim_economic_activity`: 937 rows. Primary key `activity_id`. Captures 6-digit SCIAN codes, descriptions, and broad categories (*Retail*, *Services*, *Manufacturing*).
- `dim_crime_type`: 281 rows. Primary key `crime_type_id`. Classifies legal offense categories (*Robbery*, *Homicide*, *Vehicle Theft*).

2) Fact Tables

- `fact_census_demographics`: Grain: 1 row per AGEB (2,431 rows). Foreign key `cvegeo`. Measures: total population, economically active population (PEA), and working-age population.
- `fact_economic_units`: Grain: 1 row per active business establishment (472,617 rows). Foreign keys: `cvegeo`, `activity_id`. Measures: coordinates and employment strata.
- `fact_crime_events`: Grain: 1 row per criminal investigation file (196,990 rows). Foreign keys: `cvegeo`, `crime_type_id`, `date_id`. Attributes: incident time (24-hour format) and point geometry in EPSG:6372.

C. Spatial Indexing and Materialized Aggregation

Evaluating 196,990 crime coordinates against 2,431 multipolygons requires over 478 million geometric intersection evaluations. To optimize execution, Generalized Search Tree (**GiST**) indices were deployed:

```
CREATE INDEX idx_dim_geo_geom ON dim_geography USING GIST(
    geom);
CREATE INDEX idx_crime_geom ON fact_crime_events USING GIST
    (geom);
CREATE UNIQUE INDEX idx_dim_geo_cve ON dim_geography(cvegeo
    );
```

TABLE II
OFFICIAL PROJECT KPIS: OPERATIONAL DEFINITIONS

Category	KPI Name	Operational Description
Demographic	Total Population	Aggregation of resident census count per AGEB
Demographic	Population Density	Total inhabitants divided by area in square kilometers
Demographic	PEA Rate	Economically active population divided by total population
Demographic	Working-Age Share	Population aged 15 to 64 divided by total population
Economic	Total Businesses	Count of active economic units from DENUE
Economic	Business Density	Total economic units divided by area in square kilometers
Economic	Biz per 1k Residents	Total businesses scaled per one thousand residents
Economic	Retail Density	Retail establishments divided by area in square kilometers
Economic	Service Density	Service establishments divided by area in square kilometers
Economic	Dominant Sector	Statistical mode of economic activity classification
Public Safety	Total Crime Incidents	Total criminal investigation files from FGJ
Public Safety	Crime Rate per 1k Hab	Total crimes scaled per one thousand resident inhabitants
Public Safety	Crimes per Business	Total crimes divided by active commercial units

The GiST index reduces spatial query execution from quadratic scanning to logarithmic tree traversal, allowing point-in-polygon queries to resolve in under 2.5 seconds. To accelerate downstream analysis, a materialized view (`mv_analytical_ageb_kpis`) pre-computes all 13 territorial KPIs across the 2,431 AGEBs while enforcing strict geometric validity and referential integrity.

IV. TERRITORIAL KPIS AND DENOMINATOR DIAGNOSTICS

A. Official Project KPIs

Table II defines the 13 required Key Performance Indicators specified in Section 05 of the syllabus.

B. Diagnostic Resolution: The Small Number Problem

Standard queries sorting by per-capita residential crime rate produced rankings dominated by polygons with tiny populations (such as 1, 2, 4, 7, or 68 inhabitants), yielding mathematically inflated rates exceeding 31,000 crimes per one thousand residents.

An audit confirmed that the demographic layer was populated accurately: the median AGEB population across Mexico City is 3,396 residents (mean of 3,759), with only 17 zero-population AGEBs (0.7%) and 44 AGEBs with under 50 residents (1.8%).

Investigating these outliers revealed the classic Small Number Problem (Denominator Effect) in urban criminology [6]:

- AGEB `0900700012475` (Iztapalapa): 68 residents, 807 crimes, and 6,223 businesses (*Central de Abasto CDMX*, over 500,000 daily commercial visitors).

- AGEB 0901700010598 (Venustiano Carranza): 4 residents, 124 crimes, 94 businesses (Airport logistics terminal).
- AGEB 0901600010228 (Miguel Hidalgo): 2 residents, 33 crimes, 42 businesses (Polanco commercial office strip).

Dividing crime counts by resident night-time populations induces artificial inflation in commercial zones because offenses target the daytime floating population. Two complementary solutions were implemented:

1) **Crimes per Business Ratio:** Normalizes offense volume against the commercial operating footprint by measuring total recorded criminal offenses relative to the count of active economic units.
2) **Area-Normalized Spatial Densities:** Computes occurrences per physical square kilometer (crimes per square kilometer and businesses per square kilometer), eliminating small-population distortion across all 2,431 AGEBs.

Figure 2 displays the spatial distribution of these densities classified into five equal quantile tiers. While Population Density is dispersed across the intermediate residential rings (Iztapalapa, Gustavo A. Madero), Business Density and Crime Density concentrate heavily within the central core (Cuauhtémoc, Benito Juárez, Miguel Hidalgo).

V. Spatial Econometric Analysis and Empirical Findings

A. Correlation Analysis: Pearson vs. Spearman Rigor

Statistical correlation must respect variable measurement scales [5]:

- **Pearson Linear Correlation:** Evaluates linear association between continuous interval or ratio variables. It assumes bivariate normality and is sensitive to extreme values.
- **Spearman Rank Correlation:** Evaluates monotonic association by converting continuous measurements into ordinal ranks. Because urban density distributions exhibit pronounced right-skewed tails (such as the Central de Abasto), Spearman's rank correlation serves as the primary robust measure.

As shown in Fig. 3:

- **Crime vs. Business Density:** Exhibits a strong positive linear correlation (Pearson coefficient of 0.50) and an even stronger rank correlation (Spearman coefficient of 0.58, with p-value below 0.0001).
- **Crime vs. Population Density:** Displays a substantially weaker correlation (Pearson coefficient of 0.28, Spearman coefficient of 0.36, with p-value below 0.0001).
- **Total Crimes vs. Total Businesses:** Yields a Pearson coefficient of 0.61, compared to 0.40 for total resident population.

Commercial presence is approximately twice as strong a predictor of crime concentration as residential population density.

TABLE III
Global and Bivariate Moran's I Results (999 Permutations)

Indicator	Moran's I	z-score	Pseudo p-value
Crime Density	**0.5568**	44.82	0.0010
Business Density	**0.4371**	35.19	0.0010
Bivariate: Crime vs. Neighboring Business	**0.3432**	27.64	0.0010

B. Spatial Weights Matrix

Spatial interaction is modeled using Queen Contiguity: polygons are designated as spatial neighbors if they share a common boundary edge or vertex [5]. One detached physical island (AGEB 0901200011229 in Tlalpan) was resolved by uniting the Queen contiguity graph with a four-nearest-neighbors graph. The resulting weights matrix was row-standardized so that neighbor weights sum to unity, encompassing 2,431 nodes with an average of 6.45 neighbors per AGEB.

C. Global and Bivariate Spatial Autocorrelation

Global Moran's I evaluates whether crime and business density exhibit spatial clustering versus the null hypothesis of spatial randomness, with statistical significance tested through 999 random permutations [6]. Bivariate Moran's I evaluates spatial cross-correlation by measuring the degree to which crime density in a given AGEB associates with the average business density of its surrounding contiguous neighbors.

As reported in Table III, both indicators decisively reject spatial randomness (pseudo p-values equal to 0.001). High-crime polygons cluster contiguously, and Bivariate Moran's I (0.3432) confirms that criminal activity is systematically drawn to neighborhoods characterized by elevated commercial density.

D. Local Spatial Patterns (LISA) and Hot-Spot Overlap

Local Indicators of Spatial Association (LISA) [6] identify localized spatial clusters by comparing the standardized value of each AGEB against the spatial lag of its neighbors. At a 95% confidence level, LISA categorizes spatial patterns into High-High (Hot Spots), Low-Low (Cold Spots), High-Low, and Low-High outliers.

Figure 4 displays the cluster maps alongside a synthetic Hot-Spot Overlap Map:

- **Business Hot Spots:** 107 High-High polygons concentrated in Cuauhtémoc (Centro Histórico, Roma, Condesa), Miguel Hidalgo (Polanco), and northern Benito Juárez.
- **Crime Hot Spots:** 302 High-High polygons forming a contiguous corridor across the central urban core.
- **Hot-Spot Overlap:** 98 AGEBs are simultaneously High-High for both Crime Density and Business Density—meaning that 91.6% of all business hot spots in Mexico City are also crime hot spots.

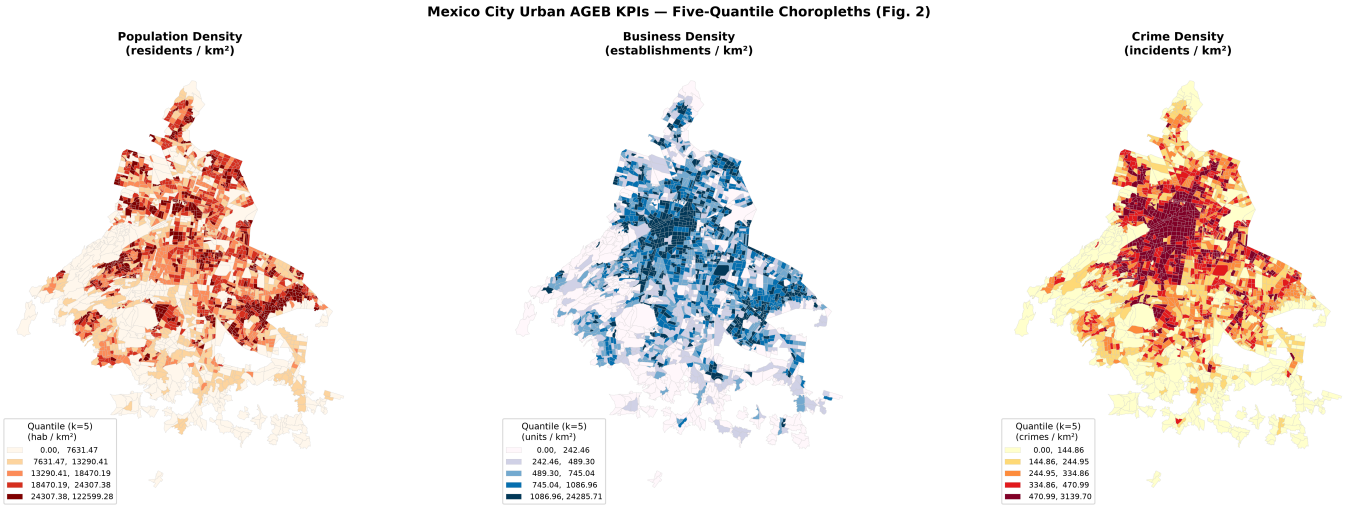


Fig. 2. Mexico City urban AGEB Key Performance Indicators classified into five equal quantile choropleth maps. Left: Population Density (inhabitants per square kilometer). Center: Business Density (units per square kilometer). Right: Crime Density (incidents per square kilometer). Spatial boundaries: 2,431 urban AGEBs, EPSG:6372.

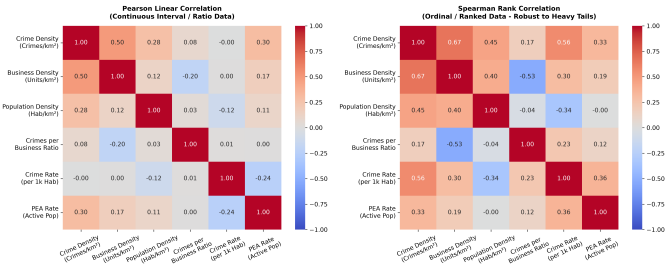


Fig. 3. Statistical correlation matrices across all 2,431 urban AGEBs in CDMX. Left: Pearson linear correlation coefficients on continuous ratio metrics. Right: Spearman rank correlation coefficients on ordinal ranked data robust to heavy right tails.

- **Cold Spots:** Low-Low clusters for both indicators concentrate in the peripheral residential zones of Iztapalapa, Tláhuac, and Xochimilco.

E. Theoretical Synthesis and Policy Implications

These findings provide empirical validation for urban criminology theories:

1) **Routine Activity Theory (Cohen and Felson):** Crime requires the convergence in space and time of motivated offenders, suitable targets, and the absence of capable guardians. Commercial corridors concentrate retail transactions, foot traffic, and parked vehicles, presenting abundant targets.
2) **Natural Surveillance Breakdown (Jane Jacobs):** While mixed-use streets promote eyes on the street, hyper-concentrated commercial corridors experience an exodus of resident guardians after business hours. The collapse of informal surveillance exposes these zones to elevated nighttime burglaries and robberies.

Policy Implications: Municipal police deployment has historically been allocated proportional to residential population quotas. This strategy starves commercial corridors of necessary patrols. Municipal authorities should dynamically reallocate tactical patrols based on Crime Density per square kilometer and Business Activity Ratios, deploying surge resources during peak commercial windows (Fridays, 12:00–18:00) rather than dispersing officers uniformly across low-crime residential peripheries.

VI. Limitations and Research Cautions

1) **Modifiable Areal Unit Problem (Openshaw):** Spatial statistics depend on the boundary definition of urban AGEBs. Aggregation to alternative partitions would alter observed correlation values.
2) **Ecological Fallacy (Robinson):** Area-level statistical associations between commercial density and crime do not imply individual causation.
3) **Reporting Bias:** FGJ registries measure reported crimes; offenses with lower reporting rates (such as extortion) are underrepresented compared to vehicle theft.

VII. Conclusion

This project successfully designed, implemented, and validated a reproducible Geospatial Data Warehouse for Mexico City at the urban AGEB grain within an automated Dockerized PostgreSQL/PostGIS environment.

The research demonstrated that traditional per-capita residential crime rates suffer from the Small Number Problem in non-residential commercial corridors. By implementing area-normalized crime densities and crime-to-business ratios across all 2,431 AGEBs, empirical findings proved that crime in Mexico City is driven overwhelmingly by commercial attractors (Pearson coefficient of 0.50, Spearman coefficient of 0.58) rather than residential population density (Pearson coefficient

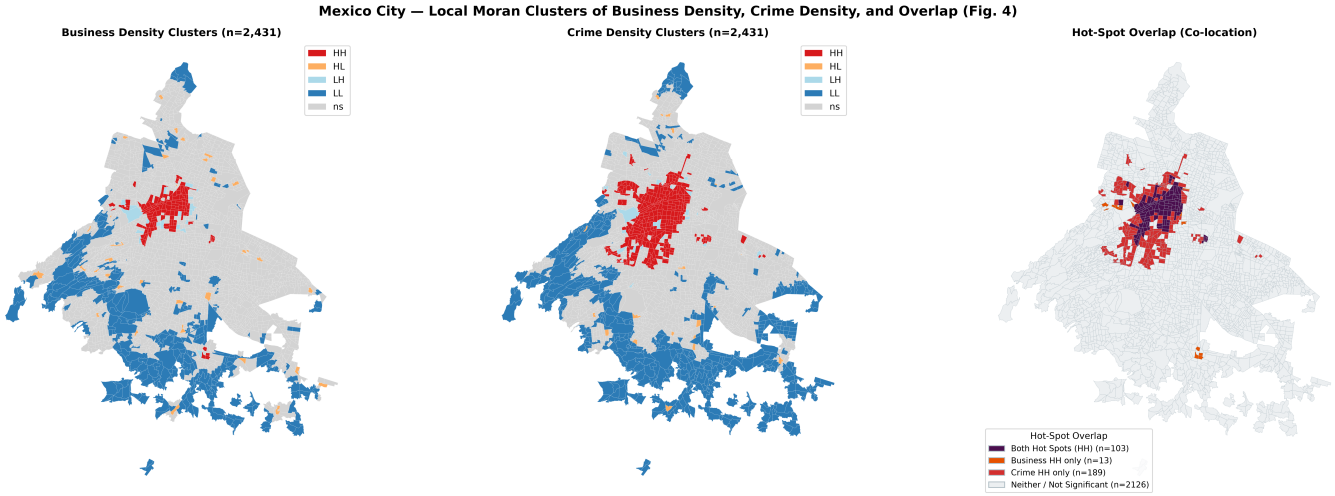


Fig. 4. Local Moran cluster analysis across 2,431 urban AGEBs in CDMX. Left: Business Density clusters. Center: Crime Density clusters. Right: Hot-Spot Overlap map displaying spatial coincidence between commercial attractors and crime clusters.

of 0.28, Spearman coefficient of 0.36). Spatial econometrics confirmed heavy spatial autocorrelation (Moran's I of 0.5568 for crime, 0.4371 for businesses), with 91.6% of all business hot spots co-locating with crime hot spots in the central urban core. The resulting platform establishes a reliable foundation for evidence-based urban analytics and public safety policy design.

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